

Software Engineering

Hands-on Session (20/08/2026)

Name: Kallepalli Sai Meghana

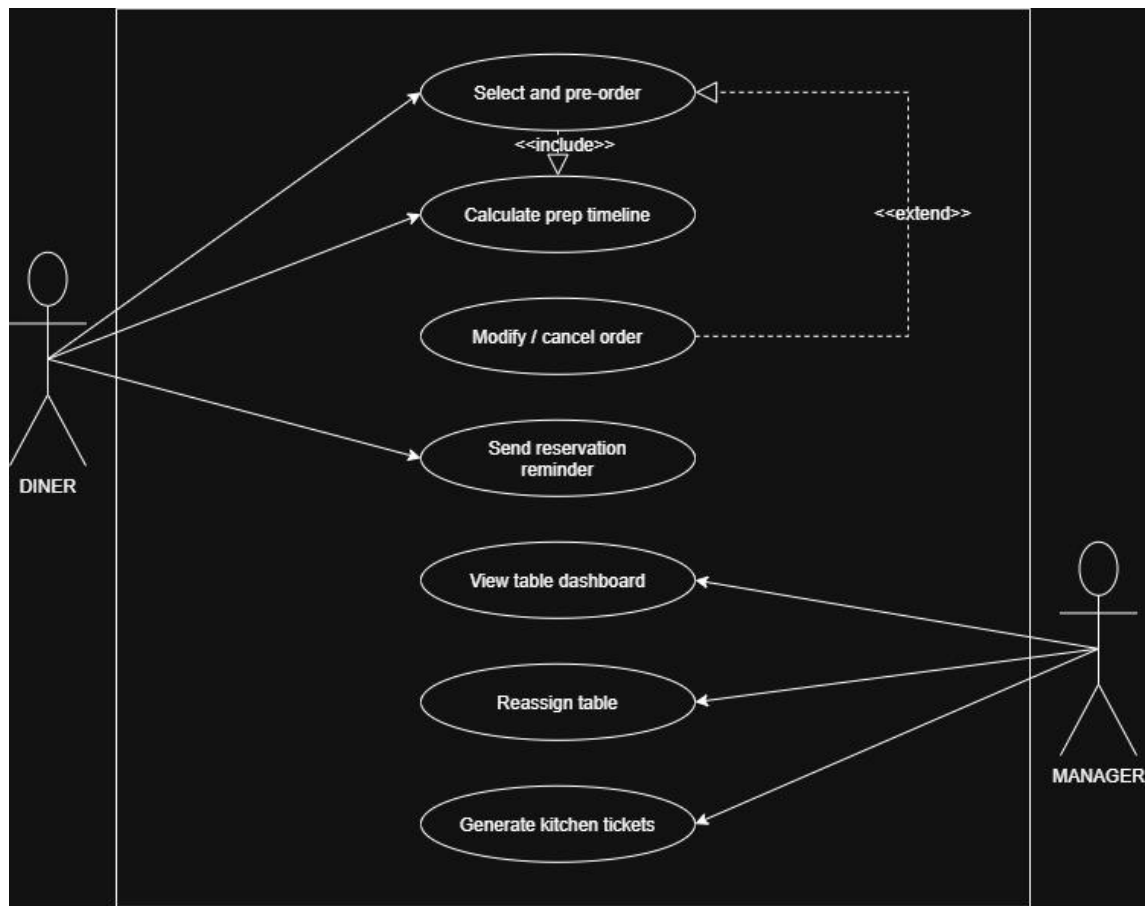
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Section: E

1.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Diner shall select an open table from an interactive 2D floor plan and link pre-ordered menu items to that reservation.	High	Pass: table locked + pre-order ticket queued. Fail: two diners reserve the same table for overlapping windows.	Replaces walk-in uncertainty with guaranteed seating + food prep.
FR-002	Functional	System shall compute an estimated "food-ready" time by combining kitchen prep time per dish with current kitchen order load, and display it to the diner as a suggested arrival time.	High	Pass: estimate shown before booking confirmation, within ± 5 min accuracy in test data. Fail: no estimate or arrival time not linked to prep time.	Prevents diners arriving too early (cold food) or too late (kitchen backlog).
FR-003	Functional	Restaurant Manager shall view a live dashboard of all tables with status (Available, Reserved, Seated, Billed) and manually override/reassign a table.	Med	Pass: status change reflects on manager terminal instantly and diner is notified if reassigned. Fail: manager cannot override a stuck/incorrect status.	Staff need an escape hatch for real-world exceptions (no-shows, walk-ins, disputes).
FR-004	Functional	System shall send an automated reminder to the diner before the reservation window and allow modification/cancellation of the pre-order up to a configurable cutoff time.	High	Pass: reminder sent, edits before cutoff update kitchen ticket. Fail: edit accepted after cutoff or no reminder sent.	Reduces no-shows and last-minute kitchen waste.
FR-005	Functional	Restaurant Manager shall generate a kitchen prep-ticket queue, auto-sorted by required-ready-time, that updates in real time as new pre-orders arrive.	Med	Pass: queue re-sorts within 1s of a new order. Fail: tickets shown in raw arrival order regardless of ready-time.	Kitchen needs actionable sequencing, not just a list of orders.
NFR-001	Performance & Security	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms, over an authenticated connection.	High	Pass: latency + auth benchmarks pass under simulated peak load. Fail: any terminal shows stale status >500ms or unauthenticated write succeeds.	Prevents double-booking and protects reservation data integrity.
NFR-002	Usability & Reliability	The floor plan view shall render within 2 seconds on a simulated 3G connection, and the booking service shall maintain 99.5% uptime during peak hours (Fri–Sun, 6–10 PM).	Med	Pass: load-time and uptime targets met in test. Fail: render >2s or downtime during peak window in test logs.	Booking is most critical exactly when load is highest

2. UML Use-case Diagram



3. Use-Case Flow Specification

Use Case: Select Table & Place Pre-Order

Actor: Diner

Preconditions:

- Diner is authenticated in the app.
- At least one table is available for the diner's chosen date/time slot.
- Restaurant's menu and kitchen prep-time data are loaded and current.

Postconditions:

- The selected table is marked "Reserved" for the diner's time window.
- A kitchen pre-order ticket is created and queued with an estimated ready time.
- A booking confirmation (table + estimated arrival time) is sent to the diner.

Main Success Scenario:

1. Diner opens the app and selects a restaurant, date, and time slot.
2. System displays the live 2D floor plan showing table availability for that slot.
3. Diner selects an open table.
4. System displays the menu; diner selects items to pre-order.
5. System calculates prep timeline («include» — invokes *Calculate Prep Timeline*) using item prep times and current kitchen load.

6. System displays a suggested arrival time and booking summary to the diner.
7. Diner confirms the booking.
8. System locks the table (status → Reserved), queues the kitchen pre-order ticket, and sends a confirmation to the diner.

Alternate Flow

Table becomes unavailable during selection also known as concurrency conflict:

- At step 3, another diner reserves the same table before this diner confirms (step 7).
- 3a. System detects the conflict when the diner attempts to confirm.
- 3b. System displays "Table no longer available" and returns the diner to the floor plan (step 2), refreshed with current availability.
- 3c. Diner selects a different table and resumes at step 3.
- Use case resumes at step 4 with the newly selected table.